Adaptive contraction beyond the October 26 SAT frontier

New coefficient bounds, a pinned review of twenty packets, and the remaining logarithmic-order barrier

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Abstract

We review all fifteen refinement packets filed under `Computability/SubexponentialSAT`, the five subsequent SAT archives in `docs/incoming`, and their common source manuscript at a pinned ProveIt snapshot. We obtain three improvements to the best explicit contraction certificates in that corpus. On the normalized scale $s/\sqrt{\log_2 s}$, where s counts Boolean wires, the polynomial-space time coefficient decreases from 55.7094783 to 46.1848456. The fastest general cached time coefficient decreases from 35.3545881 to 34.3252828, with space coefficient 34.0241780. For roots of degree at most $s^{1/12}$, the cached time coefficient decreases from 15.3919332 to 13.3796491.

The first improvement selects a separator tree from complete binary linear flags. A coset-intersection inequality makes an adaptive minimax bound rigorous. The second combines a centered entropy estimate for an adaptive three-block partition with recursive matrix-based table construction. Every recursive child evaluation is charged, and all schedules include precision, packing, degree, and stopping checks.

These are strict improvements of leading coefficients in asymptotic upper-bound functions. They do *not* replace the $n \log\log n/\sqrt{\log n}$ worst-case SAT exponent by a smaller logarithmic order, and they do not establish the corpus's conditional $n \log\log n/\log n$ target. The results are relative algorithmic refinements of the imported deformation and recovery identities, with new natural-language proofs and reproducible finite checks.

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1 Outcome and comparison standard

The initial review snapshot is ProveIt commit `fd0edd17e6a29117179b40a453c0def59165bd7c`. At that snapshot, the latest commit changing the filed SAT project was `6d3adc450615`; the three newer arrival commits were `f42479bca245`, `f898a9fde4c3`, and `fa6e904d109b`. A closing check at `dde01c4a63b822c25b8eb345299346c36f837b49` incorporated the placement commit `450252e9d86c`: the same five archives are now filed under `Research/`. All 230 filed members have exactly the Git blob identities computed from the reviewed ZIP members. The existing report contents are unchanged; two previously omitted replay scripts were restored and the placement ledger was corrected. The review therefore covers the twenty committed packets through that closing snapshot. These commits are dated October 26, 2026, Pacific time. The source preprint is vendored from `openai/math@7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. The archive's provenance ledger records full hashes and exact source paths.

1.1 Three bounds that improve the reviewed frontier

Write

$$E(s) = \frac{s}{\sqrt{\log_2 s}}.$$

All numerical coefficients in Table 1 multiply $E(s)$ in a base-two exponent, with a multiplicative polynomial in input size and an additive $o(E(s))$ exponent. The main general interface has root degree at most $\lceil s^{9/10} \rceil$; the last row has a stronger premise.

Resource / root premise	Previous time	New time	New space
General, polynomial space	55.7094783	46.1848456	polynomial
General, cached	35.3545881	34.3252828	34.0241780
Cached, degree $\le s^{1/12}$	15.3919332	13.3796491	13.2530249

Table 1: Like-for-like comparison of explicit normalized certificates. The lower-degree row cannot be substituted for the general row.

Set $\beta_r = \sqrt{(r-1)(4r+10)}/(3r)$ for powers of two. With $q = 63/100$, the exact new constants are

$$C_{\mathrm{ps}} = \frac{\pi\sqrt{2}}{\ln 2}\left(\frac{9\sqrt{80}}{8}\beta_{16} + \frac{\sqrt{120}}{64}\beta_8 + \frac{\sqrt{60}\beta_4}{256\sqrt{1-q}(1-1/(2\sqrt{q}))}\right), \tag{1}$$

$$C_{\mathrm{cache}} = \frac{\pi}{\ln 2}\sqrt{\frac{7328\log_2 9}{405}}, \qquad C_{\mathrm{cache,space}} = \frac{2\pi}{3\ln 2}\sqrt{40\log_2 9}, \tag{2}$$

$$C_{\mathrm{low}} = \frac{\pi}{\ln 2}\sqrt{\frac{203840}{23391}}, \qquad C_{\mathrm{low,space}} = \frac{\pi}{\ln 2}\sqrt{\frac{200000}{23391}}. \tag{3}$$

The decreases in the respective *time coefficients* are approximately 17.10%, 2.91%, and 13.07%. They are not empirical speedups. For example, the new and old general cached certificate functions have ratio

$$2^{-(1.02930526+o(1))E(s)} \longrightarrow 0.$$

This establishes an asymptotic separation between the displayed upper-bound functions. It does not establish a lower bound on any older implementation's actual running time.

The new general cached space coefficient is also below the previous *fastest* cached branch's 34.2748030. Other slower branches have smaller space coefficients; the full time–space frontier is not reduced to a single point.

1.2 What is proved here and what is inherited

The new arguments prove adaptive summation bounds and their schedules. They inherit the source's exact deformation, local integrality, Hermite recovery, creation-precision convention, and SAT reductions. They also inherit the reviewed asymmetric packing and full-orientation Cauchy factorization. Section 3 states the precise interface. This report does not independently reprove the entire upstream ETH result, run the upstream Lean development, or supply an end-to-end SAT solver.

The phrase *strongest established bound* in this review means a displayed bound supported by a mathematical argument in the reviewed packets, relative to their declared imports. The repository itself labels the packets external and unchecked in ProveIt. We preserve that distinction. The unproved algorithmic hypotheses of earlier packets are not promoted to established algorithms by including them in a comparison.

2 Review of the recent commits and all twenty packets

2.1 The filed project

Commit `2b925d8800cd` created the project and placed the first seven refinement packets. Commit `3e096089911f` vendored the primary OpenAI manuscript; `45233238aebf` summarized the first-round claims. Commit `6d3adc450615` placed eight second-round packets. We read the project index and the report sources and proof-status material needed to compare their claims. Table 2 records every filed packet, including proposals that do not yield a stronger unconditional rate [1].

Table 2: The fifteen filed refinement packets. Coefficients are on $E(s)$ unless a different scope is stated.

Packet	Claim and relevant limit
`cauchy_mode_improvement`	Factorial modes with weight $3b + 1 - 2\,\mathrm{popcount}(b)$; leading enumeration factor $1/\sqrt{3}$. Same logarithmic order.
`eth_contraction_improvements`	Exact taper and precision accounting, streamed extraction; coefficient 648.44 versus the analyzed source schedule's 18535.75.
`3sat_precision_refinement`	Working precision $24MD/u$ reduced to $6MD/u$ at leading order. The stronger $n \log\log n/\log n$ rate needs an unproved structured-mode summation algorithm.
`sat_density_improvement`	Density-aware exponent $\frac{n}{\sqrt{\log n}} \min\{\log(2 + m/n), \log\log n\}$. Improves intermediate densities, not unrestricted worst-case order.
`eth_density_refinement`	Overlapping density-aware result and sharper precision coefficients. Same worst-case and fixed-density orders.
`3sat_refinement`	Scalar parity lift and smaller retained incidence degree. Proposed $n(\log\log n)^{3/2}/\log n$ rate needs a missing factorization.
`3sat_asymptotic_analysis`	Scoped barriers and missing degree-preserving / linear-wire interfaces; no stronger unrestricted bound.
`eth_asymmetric_packet_balancing`	General coefficient 76.90899; low-degree 31.0641. Asymmetric retained/fresh sides are a central import here.

Packet	Claim and relevant limit
`3sat_round2_refinement`	Coefficient 84.06627; matching packing, precision gap, and two-stage schedule. Includes additional audits.
`eth_balanced_packing`	Coefficient 94.74820, polynomial space; nearly lossless padding.
`eth_high_moment`	Same 94.74820 generic coefficient by a different packing argument; smaller coefficients under stronger degree premises.
`eth_matching_refinement`	Same 94.74820 generic coefficient, from matching-potential packing.
`eth_packet_refinement`	Coefficient 95.66394; sharper pair loads and additive precision guard.
`eth_recursive_refinement`	Coefficient 133.20737; actual-domain Cauchy refinement. Does not change logarithmic order.
`eth_joint_cauchy`	Polynomial-space 319.68576; cached 225.59706 with space 196.19945. Its slope-four joint basis combines with better packing.

The project also contains the separate 49-page manuscript *A $2^{o(n)}$ time algorithm for 3-SAT*, credited to Claude. Its displayed general exponent has the same $n \log\log n/\sqrt{\log n}$ order. Its reported formal development is not published with that manuscript, so this review does not treat its formal-verification claim as a locally reproduced check.

2.2 The five newer arrivals

The pinned project README's 76.90899 minimum is not the final baseline: the five ZIPs arrived after that summary [2]. Their reports combine the filed ingredients and then improve their summation or kernels.

Incoming archive stem	Polynomial-space time	Cached time / space
`subexponential_sat_frontier`	63.85220	37.93011/33.62304
`subexponential_sat_kernel_improvements`	66.60514	44.5376/39.4959
`subsat_contraction_tree`	—	35.35459/34.27480
`sat_frontier_improvements`	63.85741	38.48450/34.27480
`sat_adaptive_contraction`	55.70948	36.75031/34.02418

Table 3: Incoming comparisons; all five filenames end in `_2026-10-26.zip`. The machine-readable ledger includes exact filenames, source commits, and the qualifications on comparison-only examples. The kernel row records comparison examples, not a claimed new global theorem.

The first three archives were added in `f42479bca245`; `sat_frontier_improvements` in `f898a9fde4c3`; and `sat_adaptive_contraction` in `fa6e904d109b`. The contraction-tree report contributes binary table construction and a coordinate/CRT bridge to field matrix multiplication. The adaptive report contributes exact pair-budget partitioning and streamed separators. The kernel-focused packets also contain measured local arithmetic improvements. Those measurements concern their particular kernels and fixtures, not complete SAT runtimes, and are not used as evidence for our new asymptotic theorems.

2.3 The comparison envelope

Taking the best compatible *proved claims* across the corpus leaves the general SAT order

$$\mathrm{poly}_k(n,m)\, 2^{O_k(n \log\log n/\sqrt{\log n})}$$

and the fixed-density order $\mathrm{poly}_{k,D}(n,m)2^{O_{k,D}(n/\sqrt{\log n})}$. The density-sensitive refinement remains available by taking the better applicable algorithm. The best displayed general polynomial-space and cached contraction coefficients before this report are the first two entries in Table 1.

The conditional $n \log\log n/\log n$ proposal is asymptotically smaller than every constant refinement here, but its needed summation procedure is absent. We have not proved that hypothesis or found a valid replacement. Consequently our result improves the supported normalized contraction envelope; it does not dominate that conditional target.

3 The common interface and its entropy geometry

3.1 Normalized tensors, arithmetic, and provenance

The input is a bipartite Boolean tensor network with at most s wires, at most polynomially many vertices, and total graph/description length $\mathcal{L}$. Its contraction is

$$Z = \sum_{x\in\{0,1\}^E} \prod_v T_v(x|_{\delta(v)}).$$

The general root degree bound is $\Delta = \left\lceil s^{9/10} \right\rceil$. Initial entries lie in $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$ with $1 \le P_0, \ell \le s$. Uniform entry evaluation uses $\mathrm{poly}(s+\mathcal{L})2^{O(s^{9/10})}$ time and polynomial space. All constants in these input premises are fixed.

The imported algorithm works in

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U,V]/(f(U), V^\ell),$$

where f is monic of degree $O(\log s)$ with irreducible reduction modulo two. Its rank and precision are polynomial in s. Tensor descriptions retain their creation precision: later calls evaluate an entry at that precision and reuse its canonical coordinate representatives. Canonical lifting is not assumed to preserve ring operations.

We import the following exact facts from the source and the reviewed refinements [3, 4, 5, 6, 7]:

1. Cauchy deformation, local integral evaluations, and tapered Hermite recovery compute the original contraction exactly modulo the requested power of two.
2. Nearly lossless asymmetric padding preserves the source/target locality and gives child wire count $(2/r + o(1))M$ at an active call with M wires and fixed alphabet r.
3. Full-orientation joint Cauchy modes have certified weights $a+1-i+4q-2\,\mathrm{popcount}(q)$, precisely one source and one target group, and the exact recursive group-factor identity described below.

The new work changes how that finite group-factor identity is summed. It does not alter its deformation, recovery, or ring semantics.

3.2 Packing and precision conditions

Let $\lambda = \lceil \log_2 s \rceil$, $b = \lceil 6\log_2 \lambda \rceil$, and

$$a_d = \sum_{j=0}^{d} u_j + (d+1)b, \qquad L_d = 2^{a_d}, \qquad Q_d = r_d^{a_d}.$$

Here u_d is the retained taper range. At a node of valuation $i < u_d$, write $D = 2^{a_d - i}$. The reviewed sharp Hermite estimate is uniformly

$$p_t \le (6 + o(1)) \frac{MD}{u_d}. \tag{4}$$

For completeness, the exact linear-taper slope may be taken as

$$\rho = \left\lceil \frac{M(2L_d - 1) + 1}{L_d(2u_d - 2 + 2^{1-u_d})} \right\rceil,$$

and a sufficient evaluation precision is $p + 6\rho D + \rho u_d(a_d + 1)$. The additive buffer makes the inherited p and final additive term lower order on the schedules below.

If R, C bound the two sides' degrees, the sufficient padding gates are

$$\frac{Q_d R}{M} = o(1), \qquad \frac{Q_d^2 C (\log s)^2}{M} = o(1), \qquad \frac{L_d Q_d^2}{M} = o(1). \tag{5}$$

At the root $R = C = \Delta$. Below it, retained tensors remain on one side, and the fresh side has degree at most $2L_{d-1}$; thus one may use $R \le \Delta$, $C \le 2L_{d-1}$ in the general setting. The separate lower-degree result tracks $\max\{\Delta, 2L_{d-1}\}$ instead. Every schedule is checked before using the nearly lossless padding conclusion, closing the depth induction.

The source estimate for inherited precision is $p = O(s\lambda^2 2^{a_{d-1}})$. Since $D \ge 2^{a_{d-1}+b+1}$, $u_d \le \lambda$, and all our leaf thresholds have $s/M \le \lambda$, the ratio $p/(MD/u_d)$ is $O(\lambda^4 2^{-b}) = o(1)$. This explains why the full precision history is compatible with (4).

3.3 Exact pair budgets and paid child factors

For a fixed alphabet r, enumerate exact weight arrays $(b_{gh})_{g \ne h}$ with nonnegative entries and sum below p_t. There are polynomially many arrays because r is fixed and p_t is polynomial in s. For one array, let $\mathcal{I}_{gh}$ be the finite list of mode exponent tuples of that exact pair weight. Discard an array if any list is empty. Define

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad K(M, u) = \frac{\pi\sqrt{2}}{\ln 2} \frac{M}{\sqrt{u}}.$$

The contribution of this array is exactly

$$\sum_{(\xi_{gh}) \in \prod_{g \ne h} \mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}). \tag{6}$$

Each Z_g entry is a genuine recursive child contraction. It is explicitly charged in every time recurrence in this report.

For any prescribed mode weights $w_1, \ldots, w_v$, the recurrence

$$D(j, z) = D(j - 1, z) + D(j, z - w_j), \qquad D(0, 0) = 1$$

with the usual zero/negative boundary conventions counts exact-weight tuples. Budgets, mode counts, and integer bit lengths are polynomial in s. This table supplies exact list sizes, rank/unrank, and polynomial-delay iteration. No counting oracle is required.

Lemma 3.1 (Weighted pair entropy)**.** *For every fixed nonnegative vector $z = (z_e)$ on the directed group pairs,*

$$\sum_e z_e h_e \le \left(\frac{\|z\|_2}{r} + o(1) \right) K(M, u). \tag{7}$$

The error is uniform over active calls and over any fixed finite family of such vectors.

Proof. We include the leading-constant calculation to make the interface explicit. There are at most $c = (2/r^2 + o(1))M/D$ modes of each index on a given directed pair. Lower their weights to $\widehat{w}_q = 1 + 4q - 2\,\mathrm{popcount}(q)$ and extend q to all nonnegative integers. Both changes only enlarge the list-generating function. The weight-counting function is $x/4 + O(\log(x+2))$, so

$$\sum_{q \ge 0} -\ln(1 - e^{-\theta \widehat{w}_q}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)) \quad (\theta \downarrow 0).$$

This follows by partial summation and $\int_0^\infty -\ln(1 - e^{-x})\,dx = \pi^2/6$. For pair e with $z_e > 0$, apply the generating-product bound at θ/z_e, multiply its logarithm by z_e, and sum over pairs. Using the total exact budget gives

$$\ln 2 \sum_e z_e h_e \le \theta p_t + \frac{c\pi^2}{24\theta} \sum_e z_e^2 + O_z(c \log^2(2/\theta)).$$

Optimize the first two terms and use (4). The optimum has scale $\theta \asymp \sqrt{u}/D$. Since the additive buffer gives $D \ge \lambda^6$ up to a fixed factor, the error is $o(M/\sqrt{u})$ uniformly. Zero weights are omitted. For a finite family, take the largest error constant. □

Corollary 3.2 (Euclidean entropy envelope)**.** *Uniformly over active calls and exact budget arrays,*

$$\|h\|_2 \le (1 + o(1))K(M, u)/r.$$

Proof. For a fixed $\varepsilon > 0$, take a finite ε-net of the nonnegative unit sphere in the fixed dimension $r(r-1)$. For nonzero h, one of its vectors has inner product with $h/\|h\|_2$ at least $1 - \varepsilon$. Apply Lemma 3.1 uniformly on that finite net. This yields $\|h\|_2 \le (1-\varepsilon)^{-1}(1 + o(1))K/r$. First take the uniform size limit, then $\varepsilon \downarrow 0$. This is precisely the asserted uniform $1 + o(1)$ statement. □

This finite-net argument matters: an estimate stated only for each fixed weight vector should not silently be used with a newly data-dependent vector without a uniformity argument. The flag proof uses a different valid route: its possible vectors already form a fixed finite family.

3.4 Why the matrix backend applies to this ring

The cached construction uses the already reviewed rational-field matrix bound $\omega_{\mathbb{Q}} < 12/5$; the primary result of Alman et al. supplies a strictly stronger bound [8]. Here is the relevant ring bridge, restated from the contraction-tree packet [7].

Fix a finite rational bilinear square-matrix algorithm with exponent strictly below $12/5$. Its base size, coefficients, and denominator primes are constants. Away from that fixed finite set of primes, the same bilinear identities hold over the prime field. Recursing one product at a time gives quadratic working storage and a bound $O(N^{12/5})$ on square matrix operations, with slack absorbing fixed overhead.

Write each entry of R_p in its specified polynomial-rank basis over $\mathbb{Z}/2^p\mathbb{Z}$, and lift coordinates to integers in $[0, 2^p)$. For matrices of dimensions d_1, d_2, d_3, an entry of an ordinary coordinate-pair product is

bounded by $d_2(2^p-1)^2$. Its bit length is $O(p+\log d_2)$, polynomial in s. Compute each such product modulo enough nonexceptional odd primes to exceed this reconstruction bound, recover the exact nonnegative integer entries by CRT, combine them using the quotient ring's integer multiplication table, and finally reduce modulo 2^p. The number and numerical magnitudes of the needed primes can be bounded polynomially in the reconstruction bit length. There are only polynomially many coordinate pairs.

For rectangular matrices, split into square blocks with side comparable to $\min(d_1,d_2,d_3)$, padding by constant factors. The resulting bit-operation bound is

$$\mathrm{poly}(s+\mathcal{L})\, d_1d_2d_3\, \min(d_1,d_2,d_3)^{-3/5}.$$

Storage is a polynomial factor times the input/output arrays. Reshaping and external sorting of polynomial-bit records add a polynomial factor. No inverse of two or of a nonunit is taken in R_p. Thus for index-length logarithms H_1,H_2,H_3 the exponent is

$$H_1+H_2+H_3-\tfrac{3}{5}\min(H_1,H_2,H_3).$$

The finite checkers accompanying this report check the new combinatorial use of this backend; they do not implement the large theoretical matrix algorithm.

4 Adaptive matrix contraction with recursively built tables

Fix an exact pair-weight allocation. Write r for the fixed number of groups, $E=r(r-1)$ for the number of directed group pairs, and $h_e=\log_2|\mathcal{I}_e|\ge 0$ for the logarithm of the exact index-list size on pair e. Allocations with an empty list contribute zero and are omitted. Put

$$K(M,u)=\frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}.$$

We use the imported weighted-entropy estimate

$$\sum_e v_e h_e \le \big(\|v\|_2/r+o(1)\big)K(M,u)\quad (v_e\ge 0 \text{ fixed}).$$

Its error is uniform over active calls and allocations for each fixed v. All constants below concern the original Boolean wire budget s and $\lambda=\lceil\log_2 s\rceil$, not the number of SAT variables.

Lemma 4.1 (Adaptive triangle bound)**.** *Partition the r groups into three blocks of prescribed positive sizes. Let the three numbers of directed pairs between blocks be v_1,v_2,v_3, and put $C=v_1+v_2+v_3$, $m=\min v_i$, and $v=\max v_i$. Choose a partition minimizing the total crossing entropy. If rectangular matrix multiplication has exponent $H_1+H_2+H_3-c\min(H_1,H_2,H_3)$, then the selected triangle can be contracted with exponent at most*

$$\left[\frac{1}{r}\sqrt{\frac{(C-cm)^2}{E}+c^2v}+o(1)\right]K(M,u).$$

Here $c=3/5$ is available from the already imported matrix exponent $12/5$ via the coordinate-and-CRT bridge over the actual coefficient ring.

Proof. Write $h = \mu\mathbf{1} + z$, where $\mu = E^{-1}\sum_e h_e$ and $z \perp \mathbf{1}$. Uniformly permuting a fixed block-size partition separates each directed edge with probability C/E. The selected partition therefore has $H_1 + H_2 + H_3 \le C\mu$. For each of its three classes,

$$H_i = v_i\mu + \sum_{e \text{ in class } i} z_e \ge m\mu - \sqrt{v}\,\|z\|_2.$$

Consequently its matrix cost is at most $(C - cm)\mu + c\sqrt{v}\,\|z\|_2$. Cauchy–Schwarz and $E\mu^2 + \|z\|_2^2 = \|h\|_2^2$ give the claimed bound by Corollary 3.2. The argument remains valid when the displayed lower bound on an individual H_i is negative. It is a lower bound used inside an upper-bound calculation, not an assertion of negative index size.

The choice is computable without transcendental comparisons: minimize $\prod_{e \text{ crossing}} |\mathcal{I}_e|$ among the finitely many partitions. The list sizes are computed by the existing exact-weight dynamic program; their bit lengths are polynomial in s. The number of partitions is a constant because r is fixed. □

The cached singleton tensors Z_g are evaluated once on all their incident indices. Every such evaluation is a paid recursive child contraction. There are $2(r-1)$ incident directed pairs, so the child-call coefficient is

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

For a block B, build its boundary table by splitting $B = A \sqcup A'$ and multiplying the two child tables, retaining all intermediate boundary indices. If $|A| = a$, $|A'| = b$, and $k = a + b$, the three reshaped matrix axes have directed-pair counts

$$2a(r-k), \qquad 2ab, \qquad 2b(r-k).$$

The existing rectangular exponent $12/5$ and weighted entropy bound the squared merge coefficient by the sum of these counts minus $21/25$ times their minimum, divided by r^2. Recursively use nearly equal binary splits. A size-k table has $2k(r-k)$ boundary pairs. Thus table construction, child evaluation, and storage are all charged, with no unit-cost child oracle.

The two schedules used below have the following complete leading ledger. The entries in the middle columns are squared coefficients before division by r^2.

r	blocks	cache	largest merge	adaptive triangle	largest boundary
9	$(3,3,3)$	16	$916/25$	$162/5$	36
15	$(5,5,5)$	28	$2548/25$	$606/7$	100

For $r = 9$, the largest merge is $1+2$ inside a triple, with axis counts $(12, 4, 24)$; its score is $40 - (21/25)4 = 916/25$. For $r = 15$, the largest merge is $2 + 3$ inside a size-five block, with counts $(40, 12, 60)$; its score is $112 - (21/25)12 = 2548/25$. All smaller merges have smaller scores, as the accompanying exact checker verifies by integer arithmetic. The triangle counts are $(18, 18, 18)$ and $(50, 50, 50)$, respectively. Lemma 4.1 gives $162/5$ and $606/7$, both strictly smaller than the largest merge score.

Therefore the local paid recurrence is

$$T_d \le \mathrm{poly}(s)\,\left(2^{(\alpha_r + o(1))K_d} T_{d+1} + 2^{(\beta_r + o(1))K_d}\right),$$

with

$$\alpha_9 = 4/9, \quad \beta_9 = \sqrt{916}/45, \qquad \alpha_{15} = \sqrt{28}/15, \quad \beta_{15} = \sqrt{2548}/75.$$

The storage coefficient is $j = 2/3$ in both cases. The coordinate-and-CRT matrix bridge, polynomial-bit records, sequential recursive products, and sorting-based reshaping are exactly the imported cached-tree interfaces. No inverse of two or of a nonunit is introduced in the coefficient ring.

4.1 Generic root degree: coefficient below 34.33

Theorem 4.2 (Generic cached refinement)**.** *Under the reviewed normalized contraction interface, including root degree at most $\lceil s^{9/10} \rceil$ and its entry, precision, and creation-cap conditions, the construction above has time and space bounds*

$$(s+\mathcal{L})^{O(1)} 2^{(C_T+o(1))s/\sqrt{\lambda}}, \qquad (s+\mathcal{L})^{O(1)} 2^{(C_S+o(1))s/\sqrt{\lambda}},$$

where

$$C_T = \frac{\pi}{\ln 2}\sqrt{\frac{7328\log_2 9}{405}} = 34.32528279728\ldots, \qquad C_S = \frac{\pi}{\ln 2}\sqrt{\frac{160\log_2 9}{9}} = 34.02417800749\ldots.$$

Thus 34.33 *and* 34.03 *are valid respective coefficients for sufficiently large s. These are ordinary mathematical bounds relative to the imported exact-contraction identities; they are not a new proof-assistant certification of the entire SAT or ETH result.*

Proof. Use $r = 9$ at every active depth. Define the integer parameters

$$J_\lambda = \max\{j \geq 0 : 9^j \leq 2^\lambda\}, \quad B = J_\lambda - \lceil\sqrt{\lambda}\rceil, \quad u_0 = \lfloor B/20 \rfloor, \quad u_d = \lfloor 3u_0/4^d \rfloor \quad (d \geq 1).$$

All statements are eventual; handle the finite initial range by the original exact algorithm. Put $b = \lceil 6\log_2 \lambda \rceil$ and $a_d = \sum_{i\leq d}(u_i + b)$, and stop at $H = \lceil s/\lambda^{3/4} \rceil$ wires. Since $\sum_{d\geq 1} 3/4^d = 1$, the total taper is at most $2u_0$.

First consider hypothetical active depths up to $\frac{3}{4}\log_4 \lambda + O(1)$. The least taper is $\lambda^{1/4+o(1)}$, hence diverges uniformly. The root symmetric packing term $Q_0^2 s^{9/10}$ and every later retained-side term $Q_d s^{9/10}$ have base-two logarithm at most

$$\lambda - \frac{\log_2 9}{10}\sqrt{\lambda} + O((\log\lambda)^2).$$

The fresh-side and rounding terms have exponent at most

$$\left(\frac{2\log_2 9 + 1}{10\log_2 9} + o(1)\right)\lambda < \frac{1}{4}\lambda$$

eventually, since $\log_2 9 > 3$. These bounds absorb $s/M \leq \lambda^{3/4}$ and all polynomial logarithmic losses. Nearly lossless padding holds uniformly, and each child has $(2/9+o(1))M < M/4$ wires. This closes the depth bootstrap. Newly created degrees have exponent at most $1/(10\log_2 9) + o(1) < 9/10$; retained original degrees are unchanged.

The buffer gives $D \geq 2^{a_{d-1}+b+1}$. The imported inherited precision bound $O(s\lambda^2 2^{a_{d-1}})$ is negligible relative to MD/u, because its ratio is at most $O(\lambda^{15/4}2^{-b}) = o(1)$. The sharp precision, entropy, and child-construction interfaces are therefore valid at every active depth. All arithmetic dimensions remain polynomial.

Write $K_0^* = (\pi\sqrt{2}/\ln 2)\sqrt{20\log_2 9}$ for the root coefficient on $s/\sqrt{\lambda}$. The tail coefficients satisfy

$$K_1^* = \frac{4}{9\sqrt{3}}K_0^*, \qquad K_d^* \leq (1+o(1))K_1^*(4/9)^{d-1} \quad (d \geq 1).$$

Unrolling the fully paid recurrence gives the maximum of the root local term $\beta_9 K_0^*$ and

$$K_0^*\left[\frac{4}{9} + \frac{4}{9\sqrt{3}}\max\left\{\beta_9, \frac{4/9}{1-4/9}\right\}\right] = K_0^*\left(\frac{4}{9} + \frac{16}{45\sqrt{3}}\right).$$

The latter coefficient is strictly below $\beta_9 K_0^*$: $\sqrt{3} > 17/10$ bounds the parenthesis by $100/153$, and $(100/153)^2 < 916/2025$. Also $\beta_9 < 4/5$, justifying the displayed maximum. This pays for every child call, including the limiting full path of child calls before a leaf.

The largest table is at the root, giving $C_S = (2/3)K_0^*$. Ancestor tables occupy a sum of sizes, not a product. Products of polynomial factors over $O(\log \lambda)$ depths contribute only $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$ to the logarithm of time. Leaf work $2^{O(H)}$, initial entry work $2^{O(s^{9/10})}$, and newly created entry work $\exp(s^{1/(10\log_2 9)+o(1)})$ are lower order, including after charging the ancestor child-call enumerations. Frozen creation-time descriptions are retained exactly as in the imported interface. Substitution of β_9 and j proves the constants. □

4.2 Root degree at most $s^{1/12}$: coefficient below 13.38

Theorem 4.3 (Low-degree cached refinement)**.** *If the same interface has root maximum degree at most $\lceil s^{1/12} \rceil$, the time and space coefficients can be replaced by*

$$C_T^{\mathrm{low}} = \frac{\pi}{\ln 2}\sqrt{\frac{203840}{23391}} = 13.37964906627\ldots, \qquad C_S^{\mathrm{low}} = \frac{\pi}{\ln 2}\sqrt{\frac{200000}{23391}} = 13.25302493301\ldots.$$

Safe eventual coefficients are 13.38 *and* 13.26.

Proof. Use $r = 15$ at every depth, $A = 113/1000$, $q = 2/25$, and

$$u_d = \lfloor A(1-q)\lambda q^d \rfloor, \qquad b = \lceil 6\log_2 \lambda \rceil, \quad a_d = \sum_{i \le d}(u_i + b), \quad H = \lceil s/\lambda^{2/3} \rceil.$$

The total leading taper fraction is at most A, with root fraction $A(1-q) = 2599/25000$. The exact integer inequality $450^{113} < 2^{1000}$ proves $A(1 + 2\log_2 15) < 1$. Since $1/12 < A$, every root, retained, fresh-side, and rounding packing condition is bounded by this same strict exponent: all old and created degrees are at most $s^{\max\{1/12,A\}+o(1)} = s^{A+o(1)}$. The gap is fixed, so buffers and polynomial logarithmic factors are absorbed.

For hypothetical depths up to $\frac{2}{3}\log_7 \lambda + O(1)$, the least taper is $\lambda^{\eta+o(1)}$ with

$$\eta = 1 - \frac{2\ln(25/2)}{3\ln 7} > 0.$$

The last strict inequality follows from $(25/2)^2 < 7^3$. Packing gives true child ratio $2/15 + o(1) < 1/7$, closing the bootstrap. The precision-buffer verification is identical, with $s/M \le \lambda^{2/3}$.

The ratio of consecutive K coefficients is $\gamma = (2/15)/\sqrt{q} = \sqrt{2}/3$. Here

$$\left(\frac{\alpha_{15}}{\beta_{15}}\right)^2 = \frac{25}{91}, \qquad \gamma + \frac{\alpha_{15}}{\beta_{15}} = \frac{\sqrt{2}}{3} + \frac{5}{\sqrt{91}} < 1.$$

For an exact rational certificate, the two terms are respectively below $472/1000$ and $525/1000$, whose sum is below one. Therefore $\alpha_{15}/(1-\gamma) < \beta_{15}$, so every level and every fully charged leaf path is dominated by the root local term. The time coefficient is $(\pi\sqrt{2}/\ln 2)\beta_{15}/\sqrt{2599/25000}$ and the space coefficient replaces β_{15} by $2/3$, yielding the displayed fractions. Leaves, entry routines, polynomial factors, and canonical creation caps are handled as above. In particular the permitted original entry-work bound $2^{O(s^{9/10})}$ is unchanged and remains lower order. □

These refinements improve explicit exponential certificates within the reviewed contraction interface. The $s/\sqrt{\log s}$ scale is unchanged. The existing SAT reductions, their wire expansion, and their density or sparsification factors must still be applied. No smaller unrestricted SAT logarithmic order or numerical domination of a separate unspecified-constant manuscript is inferred from these wire-level constants.

5 Adaptive linear flags in polynomial space

The strongest polynomial-space coefficient in the reviewed incoming packets is $55.7094783189\ldots$. Its evaluator uses a fixed balanced binary tree on sixteen groups. We improve that step by choosing a tree separately for each exact pair-budget array. The choice is deterministic, needs only polynomial space, and pays for every recursive child contraction.

5.1 The finite separator problem

Fix one exact pair-budget array. For every directed pair $e = (g, h)$, $g \neq h$, let $\mathcal{I}_e$ be its finite mode-index list and put $h_e = \log_2 |\mathcal{I}_e|$. Arrays with an empty list contribute zero. The exact graphical sum has the form

$$\sum_{(\xi_e)\in\prod_e \mathcal{I}_e} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}).$$

Each Z_g is evaluated by the prescribed child-network contraction. For a binary separator tree T, let $E(T, x)$ be the union of the directed cut-pair sets on the path from its root to leaf x. Successively conditioning on the cut indices, multiplying the two conditional sums, and streaming the outer sum is exact by finite distributivity. Its number of child-factor evaluations is at most

$$\sum_{x} \prod_{e\in E(T,x)} |\mathcal{I}_e|. \tag{8}$$

The cut sets on a path are disjoint. Indices internal to a sibling branch are summed independently and do not multiply the number of calls in the chosen branch. The assignment ranks, ring accumulators, and recursion stack have polynomial bit size.

We use the inherited weighted entropy estimate in the following notation: for any fixed nonnegative vector z on directed pairs,

$$\sum_{e} z_e h_e \leq \left(\frac{\|z\|_2}{r} + o(1)\right) K(M, u), \qquad K(M, u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}. \tag{9}$$

The error is uniform on a fixed finite family of such vectors. This is the slope-four mode entropy bound, with the sharp local precision $(6 + o(1))MD/u$ and nearly lossless asymmetric padding. It does not assume that the child factors can be evaluated at unit cost.

5.2 Linear flags and an intersection inequality

Let $r = 2^j$ with $j \geq 2$ and identify the group labels with $V = \mathbb{F}_2^j$. A complete descending linear flag

$$\mathcal{F}: \quad V = V_0 > V_1 > \cdots > V_j = \{0\}, \qquad \dim V_i = j - i,$$

defines a balanced binary tree: depth i partitions V into cosets of V_i. Write $E(\mathcal{F}, x)$ for its path cut set. There are

$$N_j = \prod_{i=1}^{j} (2^i - 1) \tag{10}$$

flags, since a d-dimensional binary vector space has $2^d - 1$ hyperplanes. Thus $N_4 = 315$ and $N_5 = 9765$. Every flag path has

$$v_r = \frac{r^2}{2} + \frac{r^2}{8} + \cdots + 2 = \frac{2(r^2 - 1)}{3} \tag{11}$$

directed pairs.

Lemma 5.1 (Common-origin intersection bound)**.** *For any flags $\mathcal{F}, \mathcal{G}$ and leaves $x, y \in V$,*

$$|E(\mathcal{F}, x) \cap E(\mathcal{G}, y)| \le |E(\mathcal{F}, 0) \cap E(\mathcal{G}, 0)|.$$

Proof. Fix a nonzero difference $d \in V$ and consider directed pairs $(a, a + d)$. Let k be the unique index with $d \in V_{k-1} \setminus V_k$. The pair belongs to $E(\mathcal{F}, x)$ exactly when $a \in x + V_{k-1}$. If W_{l-1} is the analogous subspace for $\mathcal{G}$, simultaneous membership requires

$$a \in (x + V_{k-1}) \cap (y + W_{l-1}).$$

This affine intersection is empty or a coset of $V_{k-1} \cap W_{l-1}$. Its cardinality is therefore at most the cardinality of that homogeneous intersection, attained when $x = y = 0$. Sum this inequality over $d \neq 0$. □

Lemma 5.2 (Uniform flag average)**.** *The vector $a_e = N_j^{-1} \sum_{\mathcal{F}} \mathbf{1}_{e \in E(\mathcal{F}, 0)}$ satisfies*

$$a_e = \begin{cases} 1, & \text{one endpoint of } e \text{ is zero,} \\ 2/3, & \text{both endpoints are nonzero and distinct.} \end{cases}$$

Consequently

$$\|a\|_2^2 = 2(r-1) + \frac{4}{9}(r-1)(r-2) = \frac{(r-1)(4r+10)}{9}. \tag{12}$$

Proof. Every path to zero separates every directed pair incident with zero. The group $\mathrm{GL}(j, 2)$ permutes the flags and acts transitively on ordered pairs of distinct nonzero vectors. Hence the second average value is constant. Subtracting the $2(r-1)$ incident pairs from (11) and dividing by $(r-1)(r-2)$ gives $2/3$. Counting the two types of pairs proves (12). □

Theorem 5.3 (Adaptive flag separator)**.** *For each exact pair-budget array, choose a flag minimizing $\max_x \sum_{e \in E(\mathcal{F}, x)} h_e$. Streamed evaluation with that flag uses polynomial space and obeys the paid recurrence*

$$T_d \le \mathrm{poly}(s) 2^{(b_{r_d} + o(1)) K(M_d, u_d)} T_{d+1}, \qquad b_r = \frac{1}{r} \sqrt{\frac{(r-1)(4r+10)}{9}}. \tag{13}$$

A single fixed polynomial in the input-description length and s pays for input access and arithmetic, as in the common contraction interface.

Proof. For each flag select a maximizing leaf $x_{\mathcal{F}}$ and put

$$z = \frac{1}{N_j} \sum_{\mathcal{F}} \mathbf{1}_{E(\mathcal{F}, x_{\mathcal{F}})}.$$

The minimum maximum path cost is at most its average over flags, namely $\sum_e z_e h_e$. Expanding the squared norm and applying Lemma 5.1 to every pair of flags yields

$$\|z\|_2^2 = \frac{1}{N_j^2} \sum_{\mathcal{F},\mathcal{G}} |E(\mathcal{F}, x_{\mathcal{F}}) \cap E(\mathcal{G}, x_{\mathcal{G}})| \le \frac{1}{N_j^2} \sum_{\mathcal{F},\mathcal{G}} |E(\mathcal{F}, 0) \cap E(\mathcal{G}, 0)| = \|a\|_2^2.$$

Use (9) and Lemma 5.2. The vectors z belong to a finite family determined only by the fixed group alphabet, so their data-dependent choice preserves the uniform error bound. Equation (8) contributes only a constant factor r beyond the maximum path count and pays for the real child contraction.

The flag selection requires no logarithmic approximation. For each path, form the exact integer product of its list sizes; for each flag take the largest such product and select a flag with the smallest maximum. There are only $N_j r$ products, each of polynomial bit length. Exact list indexing, flag selection, and both separator and network recursion use polynomial space when streamed depth first. □

At $r = 16$, this replaces $\sqrt{170}/16$ by $\sqrt{370/3}/16$. Keeping the incoming packet's schedule already lowers its coefficient to $47.4509449603\ldots$. We give a slightly stronger complete rational schedule next.

5.3 The integer schedule and its complete bound

Theorem 5.4 (Polynomial-space certificate)**.** *Under the common normalized contraction interface, adaptive linear flags give deterministic polynomial-space bit time*

$$(s + \mathcal{L})^{O(1)} 2^{(C_{\mathrm{flag}} + o(1)) s / \sqrt{\log_2 s}},$$

where

$$C_{\mathrm{flag}} = \frac{\pi\sqrt{2}}{\ln 2} \left(\frac{9\sqrt{80}}{8} b_{16} + \frac{\sqrt{120}}{64} b_8 + \frac{\sqrt{60}\, b_4}{256\sqrt{1-q}\,(1 - 1/(2\sqrt{q}))} \right) = 46.18484558673\ldots, \tag{14}$$

where $q = 63/100$ *and* $b_r = r^{-1}\sqrt{(r-1)(4r+10)/9}$*. In particular,* 46.19 *is a valid coefficient for all sufficiently large* s*. The result retains the common exact contraction and SAT-reduction dependencies; it does not assert a smaller logarithmic order.*

Proof. Let $\lambda = \lceil \log_2 s \rceil$, $g = \lceil \sqrt{\lambda} \rceil$, $\Lambda = \lambda - g$, and $b = \lceil 6 \log_2 \lambda \rceil$. Choose

$$\begin{aligned}
r_0 = r_1 &= 16, & u_0 = u_1 &= \left\lfloor \frac{\Lambda}{80} \right\rfloor, \\
r_2 &= 8, & u_2 &= \left\lfloor \frac{\Lambda}{120} \right\rfloor, \\
r_d &= 4 \quad (d \ge 3), & u_d &= \left\lfloor \frac{37\Lambda}{6000} \left(\frac{63}{100} \right)^{d-3} \right\rfloor \quad (d \ge 3), \\
a_d &= \sum_{q=0}^{d} u_q + (d+1)b, & H &= \left\lceil \frac{s}{\lambda} \right\rceil.
\end{aligned}$$

The cumulative retained ranges through depths one and two are at most $\Lambda/40$ and $\Lambda/30$, respectively. The final tail has sum at most $\Lambda/60$, so the total is at most $\Lambda/20$. Use direct contraction at $M \le H$ and at bounded sizes for which the parameter or admissibility checks fail. The following argument proves that all checks eventually pass.

First consider hypothetical active depths $d \le \lceil \log_{20/11} \lambda \rceil + O(1)$ and $M > H$. Their minimum taper is $\lambda^{1+\ln(63/100)/\ln(20/11)+o(1)}$, which diverges because $63/100 > 11/20$. Also $a_d \le \Lambda/20 + O((\log \lambda)^2)$. At the root, $Q = 16^{a_0}$ and the packing term $Q^2\Delta$ has logarithm at most

$$\lambda - \frac{g}{10} + O(\log \lambda), \qquad \Delta = \lceil s^{9/10} \rceil.$$

At depth one $Q = 16^{a_1}$, at depth two $Q = 8^{a_2}$, and thereafter $Q = 4^{a_d}$. The retained side has degree at most Δ, and the fresh side has degree at most $2^{a_{d-1}+1}$. The cumulative range bounds give, respectively, $4/40 = 3/30 = 2/20 = 1/10$ as the leading exponent of Q. Thus the logarithm of $Q\Delta$ is always at most $\lambda - g/10 + O((\log \lambda)^2)$. The fresh-side packing and packet-rounding exponents are at most $(9/40)\lambda + o(\lambda)$ at depth one, $(7/30)\lambda + o(\lambda)$ at depth two, and $\lambda/4 + o(\lambda)$ thereafter. Thus all three asymmetric padding gates hold uniformly: the negative $\sqrt{\lambda}$ margin absorbs the buffers, polylogarithmic factors, and $s/M \le \lambda$. Newly created degrees stay below $s^{1/20+o(1)} < \Delta$.

Padding is consequently $(1+o(1))M$. The first two child ratios are $(1+o(1))/8$, the next is $(1+o(1))/4$, and the subsequent ratio is $(1 + o(1))/2$, eventually at most $11/20$ in the tail. This validates the hypothetical depth bound and closes the induction. More sharply,

$$M_0 \le s, \quad M_1 \le (1 + o(1))\frac{s}{8}, \quad M_2 \le (1 + o(1))^2\frac{s}{64}, \quad M_d \le (1 + o(1))^d \frac{s}{256\, 2^{d-3}} \quad (d \ge 3).$$

The padding error is $O(1/\lambda)$ plus terms exponentially small in $\sqrt{\lambda}$ or a positive power of s, so its product over these $O(\log \lambda)$ levels is $1 + o(1)$.

The buffer ensures $D \ge 2^{a_{d-1}+b+1}$. The inherited precision bound $p = O(s\lambda^2 2^{a_{d-1}})$ then gives $p/(MD/u) = O(\lambda^4 2^{-b}) = o(1)$; the remaining recovery term is also lower order. This validates the uniform local precision and entropy estimates used in (13). Floors are uniformly relative $o(1)$ because the minimum taper diverges.

Summing the paid exponent in (13) gives, on the scale $s/\sqrt{\lambda}$, the coefficient

$$\frac{\pi\sqrt{2}}{\ln 2}\left(\frac{9\sqrt{80}}{8}b_{16} + \frac{\sqrt{120}}{64}b_8 + \frac{\sqrt{60}\, b_4}{256\sqrt{1-q}\,(1 - 1/(2\sqrt{q}))}\right) + o(1).$$

Indeed the first two levels contribute $\sqrt{80}b_{16}(1 + 1/8)$, and depth two contributes $\sqrt{120}b_8/64$. In the tail, $M_d/\sqrt{u_d}$ has ratio $1/(2\sqrt{q}) < 1$ and first coefficient $\sqrt{60}/(256\sqrt{1-q})$. Summing its geometric series yields (14).

Polynomial work over $O(\log \lambda)$ levels adds $O(\lambda \log \lambda)$ to the time exponent. Leaves cost $2^{O(s/\lambda)}$; original entries cost $2^{O(s^{9/10})}$; and fresh entry routines cost at most $\exp(O(2^{a_d} \log s)) = \exp(s^{1/20+o(1)})$. These are all lower order, including after the ancestor evaluation counts have been paid. Creation-time tensor descriptions remain immutable under the imported conventions, so no unaccounted recursive entry expansion occurs. All ring dimensions and precisions are polynomial, and the exact flag choices and contractions are streamed. This proves the bit-time and polynomial-space statements. □

The new coefficient is approximately 17.10% below the reviewed 55.7094783189... polynomial-space coefficient. This compares normalized upper-bound functions. It is not a runtime measurement and does not identify the wire parameter s with the number of SAT variables.

5.4 A scoped optimality statement

For a fixed $r = 2^j$, the local constant b_r is optimal for a choice among balanced binary separator trees when only the Euclidean pair-entropy constraint is used. To see this, take h proportional to the

average vector a in Lemma 5.2, with distinguished vertex zero. Every balanced tree's path to zero contains all $2(r-1)$ pairs incident with zero and has the same total size v_r. Its a-weight is therefore $2(r-1)+(2/3)(v_r-2(r-1))=\|a\|_2^2$. After normalization this matches the proved coefficient. This is a barrier only for this restricted separator model and entropy relaxation. It is not a lower bound for unbalanced separator trees, arbitrary contraction methods, or SAT algorithms.

6 Consequences for SAT and the limits of the improvement

6.1 What changes after the existing reductions

The source's arbitrary-density reduction uses a normalized wire budget $s=O_k(n\log\log n)$ and other work of lower order than the displayed worst-case exponent [3]. Applying either new general contraction algorithm retains

$$\mathrm{poly}_k(n,m)\,2^{O_k(n\log\log n/\sqrt{\log n})}.$$

The polynomial-space branch retains polynomial space. The cached branch uses subexponential space of the same scale as its time exponent. For bounded clause density, the direct equality/clause network has $O_{k,D}(n)$ wires and bounded degree, so the low-degree cached theorem applies. Its order is still $2^{O_{k,D}(n/\sqrt{\log n})}$.

These consequences should not be expressed by replacing s with n inside a numerical coefficient. One must first pay the encoding's wire expansion and normalize precision and tensor-description costs. The density-sensitive result of the earlier packets can be retained by taking the better applicable reduction [9].

6.2 An explicit bounded-density wire count

A simple count makes one translation entirely explicit. Let L be the number of literal occurrences after deleting duplicate literals within clauses and tautologies. Let d_v be the occurrence degree of variable v. Make every clause vertex a left vertex. For $d_v=1$, use a unary equality vertex on the right; for $d_v=2$, use a binary equality vertex. For $d_v\ge 3$, take an equality tree with d_v external ports and d_v-2 trivalent vertices, all provisionally on the right. Its d_v-3 internal edges each receive a left binary equality vertex. Clause-to-tree edges already have opposite colors, so they require no subdivision.

The network is bipartite and has exactly

$$M=L+2\sum_v (d_v-3)_+\le 3L\le 3km \tag{15}$$

wires and maximum degree $\max\{k,3\}$. Every consistent variable value extends uniquely through its tree and the subdivisions. Unused declared variables contribute the scalar multiplier $2^{n_{\mathrm{unused}}}$. An empty clause is handled by returning zero; an empty formula gives 2^n.

Take $s=\max\{M,n+1,2\}$, so the modulus 2^{n+1} satisfies the normalized precision premise. For fixed $k,D>0$ with $m\le Dn$, put $B_{k,D}=\max\{3kD,1\}$. The low-degree result gives the explicit conservative consequence

$$T(n,m)\le \mathrm{poly}_{k,D}(n,m)\,2^{(C_{\mathrm{low}}B_{k,D}+o(1))n/\sqrt{\log_2 n}},$$

and the corresponding space coefficient is $C_{\mathrm{low,space}}B_{k,D}$. This is a direct wiring calculation, not a claim that this encoding has the optimal wire constant. It also does not provide a comparable numerical constant for the source's general-density compression.

6.3 Why the new theorems do not change logarithmic order

At a call with M wires and taper range u, the mode-entropy scale is $M/\sqrt{u}$. The new summation arguments multiply that scale by smaller constants. Their fixed group alphabets allow $u = \Theta(\log s)$ at the top of the recursion. They do not change either dependency.

This is why a substantial decrease in a coefficient can coexist with an unchanged big-O exponent class. If the objective is specifically to replace the worst-case expression $n \log\log n/\sqrt{\log n}$ by a little-o expression, that objective remains unresolved by this iteration.

6.4 The stronger conditional proposals remain unproved

The precision-refinement packet's target $2^{O(n \log\log n/\log n)}$ requires a method summing the actual structured mode expression with only $\exp(O(M/u))$ paid recursive calls, preserving degree, precision, description size, and space. Our adaptive separators still use $\exp(O(M/\sqrt{u}))$ calls. The cached method reduces the exponent of its finite tensor summation, but still spends $\exp(\Theta(M/\sqrt{u}))$ on its certified tables in the worst envelope. Neither closes the missing interface.

The other conditional routes seek a low-degree representation after compressing pair-mode lists or a uniformly linear-wire reduction. Short binary indices alone do not preserve the bounded-degree premise of the contraction theorem: a few tensors may carry a large fraction of all the new wires. Treating their entries as free recursive oracles would hide the main computation.

7 Attempted routes to a smaller exponent

7.1 Repeated mode compression and its budget obstruction

One tempting approach is to compact the exact mode indices repeatedly. There is a useful restricted warning. Suppose a fixed number t of successive, unfactored compact-mode constructions must obey taper budgets of the form

$$u_1 \log r_1 \le A_1 \log s, \qquad u_j \log r_j \le A_j \log r_{j-1} \quad (2 \le j \le t),$$

where $r_j > 1$ and the A_j are fixed constants. Multiplication gives

$$\prod_{j=1}^{t} u_j \le \left(\prod_{j=1}^{t} A_j\right) \frac{\log s}{\log r_t}. \tag{16}$$

Thus a modeled denominator $\sqrt{\prod_j u_j}$ is still at most $O(\sqrt{\log s})$ when r_t is bounded below by a constant greater than one. Within this explicit multiplicative-budget model, repeating the operation does not yield an extra power of $\log s$.

This is an algebraic implication of the stated premises, not a lower bound for all compilers, all tensor algorithms, or SAT. The earlier factorization and asymptotic-analysis packets explain how analogous packing restrictions arise for their unfactored constructions [11, 12]. A new factorization may invalidate these premises. A growing number of stages also requires fresh uniform control of the constants and is not covered by this fixed-t statement.

7.2 Grouping parity tests before expanding incidences

The general-density encoding has total retained incidence degree $K = O(n)$ inside an incidence universe of size $N = n \operatorname{polylog} n$. Its subset indexing costs $O(K \log(1 + N/K)) = O(n \log \log n)$ wires. We examined merging parity tests into fewer groups before this expansion. Reducing the incidence universe is attractive, but a single merged group can receive essentially all the allowed Taylor or nilpotent degree. Its requested coefficient may then involve all original variables. No uniformly shrinking coefficient oracle was established in this review.

It would be incorrect to assume that oracle and count only the smaller outer network. A successful proof must either exclude such concentration, exploit a new algebraic representation, or show a decreasing measure for the coefficient computation itself. We retain this as a research question, not as a theorem.

8 Further research questions

1. **Use more structure than pair entropy.** The linear-flag theorem is sharp for balanced binary trees under its Euclidean envelope, as proved in the preceding section. Can the actual packet factors rule out its extremal entropy configuration, or permit an unbalanced or non-tree summation? This asks for a verifiable structural statement about generated tensors, not a generic black-box improvement.
2. **Remove table construction as the cached bottleneck.** The centered triangle estimate already puts the final triangle below the largest binary block merge. Further improving only the final multiplication will not lower the current local certificate. A useful next result would reduce the $1 + 2$ merge score $916/25$ at nine groups, while continuing to charge all singleton child entries.
3. **Construct the missing degree-preserving interface.** A compact mode-index network needs efficient local factors with controlled degree and immutable creation precision. Supplying that interface could change the logarithmic order. Merely showing that the index strings are short cannot.
4. **Reduce the incidence universe without an expensive oracle.** Can a grouped parity lift compute concentrated coefficient queries within a strictly decreasing size/precision measure? A positive answer would address the extra $\log \log n$ factor in the general-density reduction.
5. **Optimize a resource frontier with certified schedules.** The present algorithms select clear fixed alphabets and rational tapers. A finite dynamic program over valid alphabet changes, tree choices, and memory caps could certify additional coefficient improvements. Such optimization would refine the exponential constant unless it introduces a new dependence on u.
6. **Formalize the finite new lemmas first.** The coset-intersection lemma, the linear-flag averaging identity, and the centered triangle estimate are compact mathematical targets. Formalizing them would give reusable trusted components before attempting the much larger analytic and arithmetic integration.

9 Verification, deliverables, and integration

9.1 What the finite checks establish

The accompanying programs use only the Python standard library. Every verification gate raises an explicit exception and remains active under `python -O`. Normal and optimized runs give the same mathematical results; only elapsed-time metadata may differ.

- **Flag combinatorics.** The checker enumerates all complete flags at $r = 4, 8, 16, 32$ and verifies the path sizes and uniform common-origin average exactly. It checks 798,192 exhaustive relative-leaf intersection cases for $r \leq 16$, and 64,000 further seeded cases at $r = 32$. The latter are consistency tests, not an exhaustive proof for that size.
- **Adaptive flag evaluation.** Nine integer-weight profiles exercise exact tree selection, including an extremal profile attaining the balanced-tree entropy bound. Fifteen small contractions over $\mathbb{Z}/8\mathbb{Z}$ compare streamed conditioning against full enumeration, covering a ring with zero divisors.
- **Cached partitions.** The checker enumerates all 280 partitions of nine groups into triples, and all 126,126 partitions of fifteen groups into three blocks of five. Every undirected edge crosses exactly 210 and 90,090 of these partitions, respectively. It checks 64 arbitrary nonnegative integer-weight instances of the centered triangle estimate over all 280 nine-group partitions.
- **Exact local and global certificates.** Every merge score, boundary count, rational schedule identity, recurrence-dominance inequality, and displayed strict packing inequality is checked. Coefficient bounds use rational enclosures for π, logarithms, and square roots, independently of decimal approximations. In particular the flag constant lies in $[46.184845586730, 46.184845586731]$.
- **Independent audit.** A separate review checked the new proofs, their dependencies, and the charged recursive costs. Its exact-arithmetic spot checks and written review are included.

The finite programs do not prove the asymptotic analytic estimates by exhaustion. Those proofs are in the text. They also do not recheck the entire upstream deformation, Hermite recovery, SAT reductions, or the fast rational matrix algorithm. No new formal Lean or Rocq build, full SAT implementation, or SAT benchmark is claimed.

9.2 Reproduction

From the archive's top directory, run

```
python3 run_checks.py
python3 build_report.py
python3 verify_manifest.py
```

The first command runs both new mathematical checkers normally and under optimization, compares their results after excluding timing metadata, and writes a new verification summary. The second uses `pdflatex` twice; the third checks the delivered file hashes. For an untouched archive, run the manifest check first. Rebuilding the PDF or rerunning timed checks can change its delivered hashes without changing the mathematical content.

The complete report is `report/report.tex`, with its included section sources and compiled `report/report.pdf`. The `code/` directory contains the checkers, `results/` their outputs, `audit/` the

independent review, and `provenance/` the reviewed source identities, arrival commits, and closing relocation check. No large upstream manuscripts or duplicate input ZIPs are required in the deliverable.

9.3 Suggested repository integration

The packet can be filed under `Computability/SubexponentialSAT/Research/eth_adaptive_bounds/`. Its new interface-compatible modules are the centered adaptive triangle estimate, recursively built cached tables with the two certified schedules, and adaptive linear-flag streamed separators. The integration note identifies their imports and recommended README claims.

Preserve the resource distinctions in any repository summary. The polynomial-space coefficient is 46.1848456; the faster general cached branch is 34.3252828 time with 34.0241780 space; the 13.3796491 time coefficient requires the lower root degree. The worst-case SAT logarithmic order and the conditional status of the denominator-$\log n$ target remain unchanged.

References

[1] V. Reshetnikov, ProveIt, *Sub-exponential 3-SAT*, project index and fifteen filed research packets, snapshot `fd0edd17e6a29117179b40a453c0def59165bd7c`. https://github.com/VladimirReshetnikov/ProveIt/tree/fd0edd17e6a29117179b40a453c0def59165bd7c/Computability/SubexponentialSAT.

[2] ProveIt, five SAT research archives in `docs/incoming`, arrival commits `f42479bca245`, `f898a9fde4c3`, and `fa6e904d109b`. https://github.com/VladimirReshetnikov/ProveIt/tree/fd0edd17e6a29117179b40a453c0def59165bd7c/docs/incoming. The accompanying provenance ledger identifies each internal report.

[3] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026. Pinned source: `openai/math@7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`, especially `build/quantitative-contraction.tex` and `build/quantitative-reduction.tex`. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026.

[4] *Asymmetric packet balancing*, report and proof-status ledger in `Research/eth_asymmetric_packet_balancing/`, ProveIt snapshot listed in [1]. Used for asymmetric degree provenance and nearly lossless padding.

[5] *Joint Cauchy modes*, `Research/eth_joint_cauchy/report.tex`, same snapshot. Used for the exact slope-four Cauchy basis.

[6] *Adaptive aggregation for subexponential SAT*, `sat_adaptive_contraction_2026-10-26.zip`, internal `adaptive_sat_review.tex`, especially the pair-entropy, adaptive partition, and streamed separator sections. Arrival `fa6e904d109b`; see [2].

[7] *Cached contraction by recursive table construction*, `subsat_contraction_tree_2026-10-26.zip`, internal `report/cached_tree.tex`. Arrival `f42479bca245`. Used for binary table construction and the coordinate/CRT matrix bridge.

[8] J. Alman, R. Duan, V. Vassilevska Williams, Y. Xu, Z. Xu, and R. Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349v2, October 20, 2024; SODA 2025. https://arxiv.org/abs/2404.16349v2. Only the weaker strict inequality $\omega_{\mathbb{Q}} < 12/5$ is needed.

[9] *Density-aware SAT improvement* and *ETH density refinement*, `Research/sat_density_improvement/` and `Research/eth_density_refinement/`, same ProveIt snapshot.

[10] *3-SAT precision refinement*, `Research/3sat_precision_refinement/report.tex`, same snapshot. Its structured-mode acceleration hypothesis is explicitly unproved.

[11] *3-SAT refinement*, `Research/3sat_refinement/3sat_refinement_report.tex`, same snapshot. Scalar parity lifting, compact pair-index representations, and the missing local-factorization condition.

[12] *3-SAT asymptotic analysis*, `Research/3sat_asymptotic_analysis/`, same snapshot. Scoped barriers and missing recursive interfaces.